Haolin Sun

AI Engineer | LLM Applications & Automation

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SUMMARY

AI engineer with an M.Sc. in Computer Science who builds and ships LLM agents in production. Designed and deployed a Gemini agent on Vertex AI that raised right-party contact rates from 25% to 62% at roughly $0.10 per account. Background in deep reinforcement learning research (EUMAS 2023), RAG applications, and large-scale Python automation.

TECHNICAL SKILLS

Programming & Data: Python, SQL, PostgreSQL, Pandas, NumPy, ETL, Tableau

LLM Applications: Gemini, Google ADK, AI Agents, Tool Calling, LangChain, RAG, Prompt Engineering, Pydantic

ML & Retrieval: PyTorch, Reinforcement Learning, Model Evaluation, Embeddings, Vector Search, Milvus

Cloud & Development: Google Cloud Platform (GCP), Vertex AI, Docker, Git, Linux, LangSmith

EXPERIENCE

OpenText Jan 2024 – Present

Business Analyst – AI & Automation *Waterloo, ON*

- Independently designed, built, and deployed a Gemini-powered contact-discovery agent using Google Agent Development Kit (ADK) on Vertex AI, processing ~2,000 enterprise customer accounts per month for collections workflows.
- Implemented Google Search tool calling to find contact details across official company websites, third-party directories, and PDFs and other documents; the agent then scores and ranks every contact per account by source reliability and LLM judgment, so collectors can call the highest-priority contacts first.
- Raised the right-party contact (RPC) rate on targeted accounts from **25% to 62%**, measured on collectors' real call outcomes; improved dispute resolution rate by **45%** and contributed to **~$5M per month** in additional on-time cash collection.
- Operated the agent at ~$200 per month (about **$0.10 per account**), a small fraction of the cost of equivalent manual research.
- Built and maintained 100+ Python scripts automating data processing, reporting, and operational tasks, saving hundreds of hours of manual work annually.

York University May 2022 – Sep 2023

Research Assistant – AI & Deep Learning *Toronto, ON*

- Developed deep reinforcement learning models in Python and PyTorch that exploit reward machines in continuous action domains, benchmarked across two simulated control environments.
- Achieved **15x** the baseline benchmark score with the same number of training steps; co-authored a peer-reviewed paper at EUMAS 2023.

Jinrun Logistics, Inc. May 2018 – Aug 2018

Data Analyst Intern *Qingdao, China*

- Optimized PostgreSQL queries and database settings, improving query performance by 20%; built data-processing pipelines and time-series forecasts for logistics operations.

PROJECTS

Customer Support Knowledge Assistant *Python, LangChain, Milvus*

- Built a RAG-based customer-support assistant with LangChain and Milvus, covering document ingestion, text chunking, embeddings, and vector search.
- Implemented tool-based retrieval with structured, source-cited responses so every answer is grounded in the original documents.
- Achieved an 85% answer pass rate and 90% task completion with a 3-second average response time.

EDUCATION

York University — Master of Science in Computer Science Sep 2020 – Jun 2023

University of Ottawa — Bachelor of Science in Computer Science Sep 2016 – May 2020

PUBLICATION

Haolin Sun and Yves Lespérance. *Exploiting Reward Machines with Deep Reinforcement Learning in Continuous Action Domains*. EUMAS 2023, LNCS 14282, pp. 83–99.